

PROJECT BRIEFING FOR FACULTY MENTOR · 18 AUGUST 2026

SentinelShield

Integrity-aware statewide CCTV integration and AI watchlist platform

Prepared for **Gujarat Police Innovation Challenge 2026** (portal: sentinel.gujarat.gov.in) · Category: Students / MSME startups

Official event	Gujarat Police Innovation Challenge 2026 — CCTV integration + AI analytics; prize pool reported █37 lakh; finale = live production feeds; partners i-Hub, DA-IICT, NFSU.
Our product name	SentinelShield = CyberShield CCTV (integrity / cyber) + Sentinel-X (AI / watchlist / GIS).
Chosen official model	Hybrid: Model 1 Registry & GIS (MANDATORY) + Model 3 VMS Federation/Middleware + Model 2 Unified Viewing & Analytics. Not a rip-and-replace Central VMS (Model 4).
Budget + time (one state)	Gujarat full implementation: █38.4 Cr total (█32 Cr build + █6.4 Cr hypercare) over 36 months to statewide go-live , then 12 months stabilise (48 months programme). Student prototype still █0 / 5 weeks. Detail §8.7–8.8.
Decision asked	Approve this as the official college project / hackathon entry and the scope in Section 16.

1. Why this document exists

The official hackathon portal is organised as seven steps (Understand → Models → Solution approach → 50-camera live test → Submit → Scale to ~80,000 cameras → Evaluation). Our earlier CyberShield / Sentinel-X notes did **not** cover GIS, camera registry, VMS federation, 50-camera onboarding, watchlist matching, vehicle route history, department-wise data, cost–benefit, or statewide rollout. This briefing **fills every official box** so the mentor can see that the project is aligned to Gujarat Police’s problem — not only a college YOLO demo.

2. STEP 1 — Understand the challenge (official)

Gujarat Police want one usable picture from more than **80,000 CCTV cameras** that today sit on different departments, vendors, Video Management Systems (VMS), codecs and networks. The challenge is not “train one more detector”. It is **integration + trust + analytics + scale**.

2.1 Four key challenges from the portal — and our response

Official key challenge	What it means in Gujarat	What SentinelShield does
Heterogeneous infrastructure	Safe City, traffic, municipal, campus, highway, police thana cameras; ONVIF / proprietary / NVR-only; H.264/H.265; different time sync.	Adapter layer per source type (RTSP, ONVIF, VMS export, file, HLS). Camera registry stores vendor, codec, timezone, owner dept.
Multiple VMS & vendors	Milestone, Genetec, HikCentral, CP Plus, iVMS, custom NVRs cannot be replaced overnight.	Federation, not rip-and-replace. We pull streams or events via RTSP/ONVIF/API; existing VMS stays for recording.
Network, security & interoperability	Open RTSP, default passwords, no TLS, spoofed / looped video, no common ID for a camera.	CyberShield: SHA-256 hash chain, freeze/black/replay detect, RBAC, AES evidence, camera identity in registry.
Analytics & scalability to ~80,000 cameras	Cannot run GPU on every stream 24×7. Need watchlist + ANPR + search, not wall-of-screens.	Tiered AI: metadata-only on most cams; full infer on hotlist / event / sampled streams. Horizontal GPU workers.

Table 1. Portal Step 1 mapped to the project.

2.2 Statewide vision we accept

- One **camera registry** and map (GIS) for every on-boarded camera in Gujarat.
- Police can search a plate / person / incident across districts without logging into 20 VMS consoles.
- Live alerts when a watchlist entity appears; officer can replay a trusted clip with a hash.
- Departments keep ownership of cameras; Police get a lawful, role-based view.
- Human-in-the-loop: AI proposes, officer confirms (reduces false dispatch).

3. STEP 2 — Integration model we choose

The portal lists four reference models plus Hybrid. **Model 1 is mandatory.** We do not pick Model 4 as the only path: forcing every district onto one new VMS is politically and operationally unrealistic in a 80k mixed estate.

Portal model	Our stance	How it appears in SentinelShield
1. Registry & GIS Foundation [MANDATORY]	Must build	camera_id, lat/long, district, PS, dept, vendor, VMS, stream URL (vaulted), health, last hash. Leaflet / OpenStreetMap map + GeoJSON.
2. Unified Viewing & Analytics	Build for PoC + scale design	React mosaic of selected cameras; AI overlays; alert inbox; not a 80k live wall.
3. VMS Federation & Middleware	Primary integration strategy	Ingest adapters + message bus. Existing NVRs keep archive. We federate live + metadata.
4. Central VMS & AI Platform	Not our v1; optional long-term	Would mean replacing district VMS. We document as Phase-3 only.
Hybrid / Innovative	Official choice on the form	1 + 3 + 2, plus CyberShield integrity as the innovative layer judges usually miss.

Table 2. Step 2 — tick Hybrid; implement Model 1 as the backbone.

Innovation claim for mentors/judges: federation platforms show video; they rarely prove the frame was not looped. SentinelShield attaches a **trust score** to every camera and every alert. A stolen-car hit on a tampered feed is marked UNTRUSTED.

4. STEP 3 — Expected solution approach (all ten official tiles)

Portal text: build a deployment-ready design that continuously processes provided CCTV feeds; integrate live streams with a **searchable watchlist** (stolen vehicles, wanted / missing persons, blacklisted vehicles, suspect lists); continuous AI; automated alerts on match; full workflow (database, matching, alerting, UI); representative datasets; evaluators may ask us to find a specified vehicle or entity in the feeds in real time.

4.1 Overall architecture

Logical layers: (1) Edge / existing VMS, (2) Adapter & ingest, (3) Registry & GIS, (4) Integrity engine, (5) AI workers (ANPR, vehicle, person/face-optional, activity), (6) Watchlist matcher, (7) Fusion & case file, (8) API, (9) Operator console + GIS, (10) Audit / evidence store.

Dept cameras / VMS → Adapters (RTSP/ONVIF/API) → Ingest bus → Registry+GIS | Hash-chain | AI workers → Watchlist match → Fusion → API → React+Map → Officer

4.2 Integration strategy

- **Do not rip VMS.** Onboard via: ONVIF device service, RTSP URL from NVR, vendor API if given, or secure file/HLS drop for air-gapped sites.
- Each camera gets a UUID in the registry before the first frame is accepted (Model 1 first).
- Time: NTP policy; store both camera timestamp and server UTC (fixes vendor clock drift).
- Multi-tenant: owner department vs Police viewer vs State SOC, via roles.
- For the 50-camera test: CSV/Excel bulk onboard (IP, RTSP, lat, long, PS name) + health ping.
- Government-feed day: same adapters; credentials injected by organisers, never hard-coded.

4.3 AI and video analytics

Analytic	Hackathon / PoC	Why Police need it
Vehicle detect + track	YOLOv8 + ByteTrack	Follow a car across frames and cameras
ANPR (Indian plates)	Plate YOLO + PaddleOCR / EasyOCR; GJ, MH, RJ... patterns	Stolen / blacklisted / suspect vehicle
Watchlist person	Optional InsightFace on consented / synthetic gallery only	Wanted / missing — high ethics bar
Tamper / freeze / loop	OpenCV + SHA-256 chain	Do not chase a ghost on looped video
Cross-camera search	Plate / track-id / time-window query	Route reconstruction
Unusual activity (v2)	Out of PoC unless time	Crowd / loiter — document as roadmap

Table 3. Analytics stack. Evaluation will likely force a designated vehicle.

Matching logic: normalise plate (strip spaces, map O/0); fuzzy Levenshtein ≤ 1 for OCR noise; watchlist tables with list_type, valid_from/to, issuing_unit, priority. Hit = event + clip + GIS pin. Officer Confirm / Reject. Reject feeds active learning later.

4.4 Cybersecurity architecture (was missing in first notes)

- Identity: JWT + roles (Admin, Onboarder, Operator, Auditor, Police-view). 2FA in scale design.
- Stream secrets in vault / env, never in Git. TLS on API. Separate CCTV VLAN in scale design.
- Integrity: SHA-256 per 2–5 s segment + previous-hash chain; verify UI for any exported file.
- Tamper sensors: black, freeze, replay, sudden resolution/codec change, source-IP change.
- Evidence: AES-256-GCM export; audit log append-only (who watched, who exported).
- Privacy: DPDP Act 2023 — purpose limitation, retention, no public GitHub of real faces/plates from gov feeds.
- Hardening: no default passwords doc for field cams; ONVIF digest; disable UPnP in rollout SOP.

4.5 Deployment architecture

- **PoC / own-feed:** Docker Compose on one laptop or college PC (API, worker, DB, React, optional Redis).
- **50-camera live test:** 1× GPU box (or organiser GPU) + 1× CPU API + Postgres + object store; workers autoscale by queue depth.
- **District:** ingest close to bandwidth (district DC or police line); only metadata + alerts to State SOC.
- **State:** registry, watchlists, GIS, case search, policy. Edge does pixels; centre does decisions.
- Kubernetes later; Docker is enough to score “deployment-ready” if compose files + env samples exist.

4.6 Infrastructure sizing (indicative — for mentor, not a purchase indent)

Tier	Cameras	What we run	Rough compute
Student prototype	3–8 virtual + phone	YOLOv8n every 3rd frame, CPU	1 laptop, ■0
Live test (Step 4)	~50 heterogeneous	ANPR+track on all or on motion; 10–15 FPS detect	1× RTX 4090 / L4 class or 2× 3060; 32–64 GB RAM; 10 GbE if local
District (design)	2,000–8,000	Hotlist streams + sampled rest	GPU farm sized on <i>policy</i> , not 1 GPU/camera
State (design)	~80,000	Metadata plane + search + GIS	See Section 8; phased, not day-1 full infer

Table 4. Sizing philosophy: never promise 80k simultaneous full-HD YOLO.

4.7 Cost–benefit analysis (student honesty + police value)

Student build cost is ■0. Statewide TCO is a design estimate for judges, not our invoice. Benefit is officer-hours saved on manual rewind, faster stolen-vehicle recovery, and fewer false pursuits on tampered video.

Item	Without platform (today)	With SentinelShield (target)
Find one vehicle across cities	Hours–days; many VMS logins; CD transport	Seconds–minutes; plate search + GIS path
Trust in clip for investigation	Often none; easy loop attack	Hash chain + tamper flag; NFSU-friendly pack
New VMS licence for 80k	Very high if Model 4 forced	Reuse existing VMS; pay for middleware + GPU only
Student / college cash	—	■0 prototype; optional ■15k cams after win
Indicative state software+GPU (design only)	Fragmented spend + risk of ■200–400 Cr VMS rip-out	See §8.7: PoC ■22 L · pilot ■1.6 Cr · 4-yr state ■32 Cr · ≈ ■4,000 / camera

Table 5. Cost–benefit narrative for Step 3 tile and Step 6.

4.8 Department-wise information requirements

Department / unit	What they own / give us	What they consume
Gujarat Police (CID / Control / Traffic)	Watchlists, FIR-linked plates, live test credentials, SOP for confirm/reject	Alerts, GIS tracks, evidence pack, audit
Home / Safe City / Smart City SPVs	Camera inventory, GIS coordinates, VMS access	Health of their estate, shared incidents
RTO / Transport	Stolen / blacklisted vehicle feeds (API or file)	Match confirmations
Municipal corp. / ULBs	Ward cameras, junctions	Limited role; not crime dossiers
Highways / R&B / GSRDC	Highway CCTV, chainage + GIS	Vehicle route on corridor
Campus / jail / court complexes	Closed networks; maybe file drop only	Local SOC view
Forensics (NFSU partner)	Hash / chain-of-custody guidance	Export standard

Department / unit	What they own / give us	What they consume
i-Hub / mentor college	GPU hours, ethics review	Publication / PoC

Table 6. Step 3 — department-wise information. Watchlists never sit in the student Git repo.

4.9 Scalability strategy (preview; detail in Step 6)

- Scale metadata and registry first (cheap), pixels second (expensive).
- AI only on: watchlist-proximate cameras, motion, junction samples, officer-requested rewind.
- Shard ingest by district; one logical watchlist (replicated).
- Object storage lifecycle: hot 7–30 days clips for alerts; hashes kept years; raw video stays on existing NVR.

4.10 Future roadmap

- P0 Prototype (now): registry+GIS mock, 4–8 feeds, hash, ANPR, alerts, UI.
- P1 50-cam live test: bulk onboard, designated vehicle, route history.
- P2 District pilot: one city / one zone, real SOP, 2FA, DR drill.
- P3 Multi-district; optional face if legal cell approves; activity analytics.
- P4 Statewide registry complete; Model 4 only if Home Dept. mandates a single VMS later.

5. Data model (watchlist + matching — evaluators will look here)

Table	Key fields
cameras	id, name, district, police_station, dept, vendor, vms, lat, lon, status, trust_score, onboarded_at
streams	camera_id, protocol, endpoint_ref, codec, fps, last_frame_utc
hash_chain	camera_id, t_start, t_end, sha256, prev_sha256, ok
watchlist_vehicle	plate_norm, list_type (stolen/black/suspect), case_ref, priority, valid_to, issuing_unit
watchlist_person	temp_id, list_type (wanted/missing), embedding_ref, legal_basis, valid_to
detections	camera_id, ts, class, plate_ocr, track_id, conf, bbox, clip_uri
matches	detection_id, watchlist_id, score, status (new/ack/confirmed/false)
tracks	entity_key, points[] (cam, ts, lat, lon), route_geojson
alerts	severity, type, camera_id, match_id, trust, notified_roles
audit	who, action, object, ts, ip — append-only

Table 7. Minimum schema so “searchable database of watchlist records” is real.

6. STEP 4 — Technical evaluation / live test case (~50 cameras)

Portal live challenge: onboard ~50 heterogeneous cameras, integrate live feeds, track a designated vehicle, generate alerts and history, visualise on GIS, show route / movement history and searchable events.

Official tile	How we will pass it	Demo script
Onboard ~50 cameras	CSV import → registry. Mix: RTSP, file-loop “broken vendor”, HTTP phone, missing-GPS row (shows validation).	Import 50 rows in < 2 min; 48 green, 2 red with reason.
Integrate live feeds	Workers subscribe; mosaic shows 6–12 live, rest as health dots on map (cannot show 50 HD tiles).	Open 4 live + map of 50 heartbeats.
Track vehicle	ANPR + track_id; if plate unreadable, manual lock on bbox.	Organiser plate X → first hit in seconds.
Alerts & history	Watchlist hit → CRITICAL; operator ack; timeline of all hits.	Show inbox + filter by plate/date.
Visualise on GIS	OpenStreetMap / Leaflet; camera markers; hit pulses; district boundary GeoJSON (Gujarat).	Zoom Ahmedabad / Surat / highway as data allows.
Route, movement, search	Polyline of hit cameras in time order; table of searchable events (plate, cam, ts, trust).	Play “vehicle went Cam12 → Cam7 → Cam41”.

Table 8. Step 4 acceptance tests we will rehearse before the live window.

If organisers give only a subset of 50 as truly live, we still onboard 50 records (registry completeness) and mark offline cameras honestly — that is better than fake green badges.

7. STEP 5 — Prepare and submit (official deliverables)

Portal deliverable	Our artefact	Owner
Solution presentation	12–15 slides: problem, hybrid model, architecture, live test, scale, ethics, ■0 PoC	Pitch lead
High-Level Design document	This PDF expanded + diagrams (C4 context, sequence for match, ER)	Architect
Own-feed demonstration	Laptop demo: files + phone RTSP + tamper button + watchlist plate we filmed	ML + UI
Government-feed demonstration	Same binary; env file for official RTSP; no code change	Backend
Video & output report	5–8 min unlisted video + 8–10 page output (metrics, screenshots, limitations)	All
Submission links	GitHub (private if required), demo URL / recording, drive of HLD + PPT	TL

Table 9. Nothing on the portal submit page is left blank.

8. STEP 6 — Plan for scale (~80,000 cameras across Gujarat)

Judges score “PoC readiness” on whether we thought like a state, not whether we rented 80k licences. Numbers below are order-of-magnitude for discussion with the mentor and i-Hub, not a tender.

8.1 Hardware and software requirements (state design)

- Software: Linux, Kubernetes (or first Docker Swarm), Postgres + PostGIS, Redis/NATS, MinIO/S3, FastAPI, React, OpenSearch for events, Keycloak.
- AI: NVIDIA L4/L40S class in district GPU rooms; Triton or Ultralytics server; CPU nodes for hash + orchestration.
- Client: browser only for operators; no thick VMS replace.
- Estimate philosophy: ~5–15% of cameras under continuous heavy AI (junctions, borders, watchlist geofence); rest health + on-demand.
- $80,000 \times 0.10 = 8,000$ heavy streams. At ~15–25 streams/GPU depending on model = **order of 300–500 GPUs statewide** if naive; with motion gating and 5 FPS metadata, this can drop by 3–5×. We will present a range, not a fake exact count.

8.2 Network and bandwidth planning

- Do not backhaul 80k HD streams to Gandhinagar. Keep pixels in district / city DC.
- State WAN carries: alerts, thumbnails, hashes, search queries, occasional clip fetch.
- Rough: 2 Mbps effective after encode $\times 8,000$ heavy = 16 Gbps if all centralised — unacceptable. Hence edge infer.
- Camera onboard checklist: multicast/unicast, NAT, firewall allow-list, QoS for RTSP.
- PoC/50-cam: single LAN or organiser VPN is enough.

8.3 Storage and retention

- Raw NVR video remains with the owning department (30–90 days typical today).
- Platform stores: event clips (e.g. 30–120 s) + thumbnails + hashes + metadata.
- Alert clips: 180–365 days (policy). Hashes and audit: 7 years class (align with evidence rules / mentor+NFSU).
- PII minimisation: plate crops stored; full face gallery only with legal basis.

8.4 AI processing capacity

- Queue + priority: watchlist geofence > officer rewind > random sample quality audit.
- Model zoo: YOLOv8n/s vehicle, plate, optional face; INT8 where accuracy holds.
- SLA target (design): watchlist match alert < 5–10 s after the vehicle is readable in frame.
- Human confirm before external SMS/dispatch in v1.

8.5 Disaster recovery

- RPO / RTO (design): registry+watchlist RPO 5–15 min (stream replication); RTO 1–4 h for API.
- Active-passive State SOC DB; district ingest can run islanded (local alerts) if WAN dies.
- Daily backup of hash chain and watchlists offline.
- Chaos test in PoC: kill one worker; cameras stay green/red honestly; no silent drop.

8.6 Statewide rollout plan

Wave	Scope	Success gate
0 Hackathon	Own feeds + 50-cam test	Designated vehicle found + GIS route

Wave	Scope	Success gate
1 One commissionerate	Ahmedabad or Surat zone, 500–2,000 cams	SOP live 30 days; false-alert rate logged
2 Traffic + highway spine	ANPR-heavy corridors	RTO stolen list auto-ingest
3 All commissionerate / range	Registry 100% of known cams even if not AI-on	Map complete; health SLA
4 Remaining districts / ULBs	Adapters for leftover vendors	80k identities in registry

Table 10. Rollout: registry coverage first, AI second.

8.7 Budget plan for scale (the number judges and mentors will remember)

This is a **government-style indicative DPR sketch** for Step 6 (Hardware, Network, Storage, AI, DR, Rollout). It is *not* a student purchase indent and *not* a vendor quotation. Figures are 2026 India street/DC order-of-magnitude in INR, GST extra. They exist so the proposal looks funded-and-phased, not “we will use Colab forever”.

STUDENT NOW ■0	HACKATHON PoC ■0–0.15 L	STATE PoC (50–500 cam) ■18–26 Lakh	1 COMMISSIONERATE ■1.4–1.9 Cr	GUJARAT 4-YEAR ■28–38 Cr	AVOIDED (Model 4 VMS) ■200–400 Cr+
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Figure A. One-glance budget. Federation + reuse of existing NVRs is the cost story. Replacing 80,000 VMS licences is the bill we refuse to write.

8.7.1 Phased budget (CAPEX + first-year OPEX)

Wave / what Police buy	Cameras in play	CAPEX (indicative)	Year-1 OPEX	Who pays
W0 College / Stage-1 software	4–8 own + files	■0	■0	Team laptops
W0b Optional booth cams + SSD	2 cheap ONVIF	■0.12–0.18 L	■0	College / SSIP if wanted
W1 Official 50-cam live test	~50 gov feeds	■0 (use organiser GPU/VPN)	■0	GPIC lab
W2 State technical PoC (after selection)	50–500 mixed	■22 L (band ■18–26 L)	■4–6 L	i-Hub / challenge / Home IT
W3 One commissionerate pilot	1,500–3,000	■1.60 Cr (band ■1.4–1.9 Cr)	■28–36 L	Commissionerate + State
W4 Traffic + highway ANPR spine	+5,000–8,000 hot	■4.5–6.0 Cr incremental	■80 L–1.1 Cr	Traffic / PWD / GSRDC share
W5 Statewide federation (4 years)	Registry 80,000; AI on ~8–12k hot	■32 Cr total 4-yr (band ■28–38 Cr)	■6–8 Cr / yr at full run	Home Dept. phased outlay

Table 10A. Phased budget — small cheques first, statewide only after gates in Table 10.

8.7.2 Where ■22 lakh PoC money goes (attractive because it is specific)

PoC line item (50–500 cameras, 90 days)	Qty / note	■ Lakh
GPU inference server (NVIDIA L40S 48 GB or 2× L4 24 GB)	1 node, 3-year warranty	9.5
CPU / API / DB node (32–64 core, 256 GB, 10 GbE)	1	3.2
All-flash / NAS for 30-day event clips + hashes (~80–120 TB raw-class)	1	3.8
Firewall + jump host + vault + certificates	shared / virtual	1.2
Integration, onboarding 50–500 streams, SOP, 2-week on-site	team time monetised	2.8
Training 20 operators + documentation (Gujarati/English)	—	0.6
Contingency 8%	spares, extra NIC, HDD fail	0.9
PoC total (mid)	Ready for designated-vehicle SLA	22.0

Table 10B. A judge can audit this list. We are not asking the college for it.

8.7.3 One-commissionerate pilot — ■1.60 Cr mid case

Pilot line (e.g. Surat or Ahmedabad zone)	■ Cr
GPU farm (8–12× L4/L40S equivalent) + racks, PDU, 20 kVA UPS share	0.72
K8s / API / PostGIS / OpenSearch / MinIO cluster (3-node HA)	0.22

Pilot line (e.g. Surat or Ahmedabad zone)	■ Cr
Network (ToR, 25/40 Gb spine share, VPN concentrator)	0.10
SOC workstations (8) + video wall share + 2FA tokens	0.08
SI / integration 5 months, adapters for 4–6 VMS vendors	0.28
Training, change management, 24×7 first 90 days hypercare	0.08
AMC + onsite spares Year-0 prepaid slice	0.06
Contingency ~8%	0.06
Pilot total (mid)	1.60

Table 10C. Commissionerate cheque — still < 1% of a rip-and-replace VMS programme.

8.7.4 Four-year statewide TCO (federation) vs the bill we avoid

4-year cost head (indicative)	SentinelShield federation	Model 4: new central VMS on 80k
Reuse existing NVR / cameras / dark fibre	■0 extra (already paid by depts)	Often forklift + dual-run 18 months
VMS / channel licences	■0–2 Cr (only connectors)	■120–280 Cr (■15–35k / channel class)
GPU + edge infer rooms (8–12 sites)	■10–14 Cr	■12–18 Cr (still needed for AI)
State SOC HA + GIS + registry + search	■2.5–3.5 Cr	■4–8 Cr (bigger video backhaul)
WAN / backhaul of pixels to one city	Low (metadata + clips)	■30–60 Cr+ if naive HD centralise
SI, training, AMC, people (4 yr)	■12–16 Cr	■25–40 Cr
4-year envelope	■28–38 Cr (use ■32 Cr mid)	■200–400 Cr+ typical class
Cost per registered camera (4 yr)	≈ ■4,000 (■32 Cr / 80,000)	≈ ■25,000–50,000+
Cost per AI-hot camera	≈ ■32,000 if 10k hot streams	Still high + licence stack

Table 10D. Headline for mentor and jury: **same 80,000-camera identity + watchlist AI at roughly one-tenth the cost of buying a new statewide VMS. That is why we chose Hybrid Model 1+3+2.**

8.7.5 Annual OPEX at full Gujarat run (steady state)

Yearly operating item	■ Cr / year	Note
Power + cooling (GPU rooms, ~0.4–0.7 MW peak class if naive; gated lower)	1.4–2.2	Edge gating is a cost control, not only an AI trick
AMC 18–22% on hardware + infra software	1.8–2.4	4-hour GPU spare SLA in 3 cities
Platform SRE + SOC analysts (12–20 people loaded)	2.0–2.8	Police staff + one SI bench
Model refresh, red-team, NFSU audit, DPDP DPO support	0.4–0.6	Yearly accuracy + ethics
Cloud burst / DR drills / tape of hash chain	0.2–0.4	Mostly on-prem
Steady OPEX mid	≈ ■6.8 Cr / year	After Wave 5; not in student year

Table 10E. OPEX is honest — GPUs consume power. Gating 90% of cameras keeps this payable.

8.7.6 Benefit / ROI (why the budget is attractive, not just cheaper)

Benefit lever	Conservative annual value (indicative)
Officer time: 400 investigators save 25 min/day on rewind / VMS hopping	≈ 1.7 lakh person-hours / yr → ■25–40 Cr loaded-cost class
Faster stolen / wanted vehicle hit (Traffic + CID)	Even 50 extra recoveries/yr can exceed the ■22 L PoC many times over
Avoided false pursuit on looped / frozen camera	One avoided incident (fuel, risk, legal) justifies integrity module
Avoided dual VMS licence + training lock-in	■150 Cr+ option value vs Model 4 (Table 10D)
Reusable registry for future 112 / ICCV / disaster apps	One GIS identity for every camera — platform, not a point tool

Table 10F. Payback on the ■22 lakh PoC is designed to be **inside one quarter** if the live test already finds designated vehicles. Statewide ■32 Cr is a **multi-year insurance policy** against a ■200 Cr+ rip-out.

8.7.7 Funding map (so mentor sees we are not asking the college for ■32 Cr)

Need	Amount	Natural funder
Software + report (now)	■0	Students; OSS
Optional 2 cameras	■15,000	SSIP / department imprest / skip
Stage-1 → finale GPU hours	in-kind	GPIC / i-Hub / NFSU / DA-IICT lab
■22 L PoC kit	■18–26 L	Challenge prize (■37 L pool exists) + i-Hub seed
■1.6 Cr pilot	1.4–1.9 Cr	Commissionerate modernisation / Safe City leftover / M-SIP
■32 Cr / 4 yr state	phased GRs	Home Department + GoI Safe City / Nirbhaya-class / state IT

Table 10G. Prize money can literally buy the first real PoC server. College cash stays ■0.

Mentor one-liner: we spend nothing to compete; if Gujarat adopts the design, they buy a **■38.4 crore / 36-month statewide go-live** (plus one year stabilise) instead of a **■200–400 crore VMS replacement**, at about **■4,800 per camera all-in** — not a new camera estate.

8.8 ONE STATE — total cost and time to implement (Gujarat)

This section answers the two questions a DGP / mentor / jury will ask in one breath: **How much for the whole state?** and **How many months until it is live?** Scope = Gujarat only (~80,000 already-installed cameras). No new camera purchase is included (cameras are already a sunk departmental cost).

TOTAL COST TO IMPLEMENT IN ONE STATE (GUJARAT) ■ 38.4 crore Build ■32.0 Cr + first-year statewide hypercare / dual-run ■6.4 Cr Band: ■34–44 Cr Excludes GST, land, new cameras, existing NVR refresh	TIME TO IMPLEMENT IN THE STATE 36 months to go-live Month 0–6 PoC+pilot · 7–18 spine · 19–36 all districts Then months 37–48 stabilise / AMC. First police value at month 6 (one city).
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Figure B. The only two numbers that must appear on a slide: ■38.4 Cr and 36 months.

8.8.1 Total state cost — build-up (nothing hidden)

Cost block (Gujarat, one-time + Year-0/1)	■ Cr	% of ■38.4 Cr	What it buys
A State PoC kit (folded into B; shown for traceability)	(0.22)	—	Table 10B — not added twice
B First commissionerate pilot (includes A)	1.80	4.7%	Table 10C + buffer — live SOP
C Remaining district / range GPU + HA rooms	13.50	35.2%	Edge infer, no HD backhaul
D State SOC, PostGIS registry, search, IAM, DR site	3.80	9.9%	Gandhinagar + DR in a 2nd city
E Connectors to all major VMS / ONVIF / file-drop	2.80	7.3%	Federation, not new VMS licences
F SI, 80k registry clean-up, GIS gap survey	5.50	14.3%	Largest non-hardware work package
G Training 800+ operators, Gujarati SOP, 112 link	1.60	4.2%	Adoption, not only software
H Security audit, NFSU evidence profile, DPDP	0.80	2.1%	Trust module + legal
I Contingency on B–H	2.20	5.7%	Vendor slip / extra adapter
BUILD subtotal (B+C+D+E+F+G+H+I)	32.00	83.3%	1.80+13.50+3.80+2.80+5.50+1.60+0.80+2.20
J Statewide hypercare / dual-run for 12 months	6.40	16.7%	People + AMC + power after go-live
TOTAL TO IMPLEMENT ONE STATE	38.40	100%	■32 Cr build + ■6.4 Cr Year-1 hold
Optional add (not required)	—	—	New cameras, 100% 24×7 AI on all 80k, public 5G backhaul

Table 10H. **Quote ■38.4 crore** as “total cost to implement Gujarat”. **Quote ■32 crore** if they ask only for build CAPEX. **Quote ■6.8 Cr/year** (Table 10E) as run-cost after month 48.

8.8.2 Unit economics for one state

Metric	Value
Cameras in scope (existing Gujarat estate)	~80,000 (no camera CAPEX in this total)
All-in implementation cost / camera (■38.4 Cr ÷ 80,000)	■4,800

Metric	Value
Build-only / camera (■32 Cr ÷ 80,000)	■4,000
Steady run / camera / year (■6.8 Cr ÷ 80,000)	≈ ■850 / year
AI-hot streams at go-live (10–15% policy)	8,000–12,000 cameras under heavy infer
Cost vs new statewide VMS (■250 Cr mid-class)	~6.5× cheaper to implement
GST, inflation buffer if tender slips 12 months	Add 12–18% in a real DPR; still ■ Model 4

Table 10I. Per-camera story: less than the cost of one indoor dome, for software + AI + integrity on a camera they already own.

8.8.3 Time to implement in the state — 36-month critical path

Clock starts the day Home Department issues a work order (not from today). Hackathon / student work is **Month –3 to 0** and is free. Police start getting operational value at **Month 6** (one commissionerate), not only at Month 36.

Month	Phase	What is live for Police	Gate to next phase
–3 to 0	Hackathon / HLD (■0)	Own-feed + 50-cam test demo	Selection / in-principle nod
1 – 3	Mobilise + PoC kit	Official 50–200 streams, watchlist test	Designated-vehicle SLA met
4 – 6	First commissionerate	First city live (1.5–3k cams), SOP, 24×7 desk	False-alert rate in band; DGP review
7 – 12	Traffic + highway spine	ANPR corridors; RTO stolen list auto	Plate hit used in real cases
13 – 18	All commissionerates / ranges	Major cities on map; shared watchlist	Registry ≥ 40k identities
19 – 27	Remaining districts + ULBs	Adapters for leftover vendors; file-drop sites	Registry ≥ 70k
28 – 36	Statewide go-live	80k in registry ; AI on hot set; DR drill passed	SOC sign-off + NFSU pack used in 1 case
37 – 48	Stabilise (not “build”)	AMC, tune models, island-mode drills	Hand to steady OPEX ■6.8 Cr/yr

Table 10J. 36 months to statewide go-live. Value starts at month 6. Full estate identity at month 36.

8.8.4 Why not faster / why not slower

- **Cannot honestly finish a state in 6–9 months:** 80,000 rows of dirty GIS, 10+ VMS vendors, district WAN, and training dominate — not Python.
- **Should not take 6–7 years:** federation reuses NVRs; we are not retendering every camera.
- **Crash programme (24 months)** is possible if two commissionerates start in parallel and GIS survey is already clean — add ~15% cost (overtime / dual SI). We do **not** promise 24 months as the base bid.
- **People on the clock (indicative):** Month 1–6: ~15 engineers + 8 police SMEs. Month 7–36: ~25–35 SI + 12–20 police SOC. Not a 200-person army.
- Student team time to a portal-ready prototype remains **5 weeks part-time** (Table 14) — that is not the state implementation clock.

8.8.5 Copy-paste answers for mentor / jury

Question	Answer we give
Total cost to implement in one state?	■38.4 crore for Gujarat (build ■32 Cr + Year-1 hypercare ■6.4 Cr). Band ■34–44 Cr. No new cameras.
Time to implement in the state?	36 months to statewide go-live from work order; first city live in 6 months ; 12 more months to stabilise.
When do police get value?	Month 6 — one commissionerate on watchlist + GIS. Not only at the end.
Cost if we only do software, no hardware?	Not a serious state bid. Hardware is ~40% of ■32 Cr. Software-only is the ■0 student demo.
Can we replicate to another state?	Same software; new GIS + adapters. Roughly 70–80% of ■38 Cr, ~30 months, if camera count is similar.

Table 10K. Memorise the bold cells. Do not invent a new number on stage.

9. STEP 7 — How we will be evaluated (score against official tiles)

Official dimension	What we will show
Successful test case	Table 8 checklist signed off in rehearsal video.
PPT / PDF presentation	Mentor PDF (this) + 12-slide deck.
Solution architecture	Hybrid 1+3+2, C4 + sequence of a watchlist hit.
Working demonstration	Own-feed always-on backup if gov stream fails.

Official dimension	What we will show
Analytics quality	Plate read on our Gujarat-format clips; declare OCR accuracy on that set; no fake 99%.
Scalability & PoC readiness	Section 8 numbers + Docker + CSV onboard of 50.
Bonus consideration	Integrity/trust score; UNTRUSTED alert; NFSU-style evidence pack; DPDP note.

Table 11. Evaluation mapping — bonus is our differentiator.

10. Money — student cash is still ₹0; scale budget lives in \$8.7

Do not mix the two envelopes. **College / team envelope** = ₹0 (optional ₹15,000 booth). **Government envelope** is the attractive Step-6 plan: ₹22 lakh PoC → ₹1.6 Cr pilot → ₹32 Cr / 4 years, versus ₹200–400 Cr if Gujarat buys a new 80k-channel VMS.

Envelope	Cash	Notes
Now → Stage-1 submission	₹0	Laptops, OSS, phone, self-filmed plates, OSM maps
Optional expo cameras	₹0–15,000	Skip unless mentor wants a physical booth
Finale 50-cam live	₹0 expected	Organiser feeds + GPU
Post-win State PoC kit	₹18–26 L (mid ₹22 L)	Table 10B — prize / i-Hub, not college
Commissionerate pilot	₹1.4–1.9 Cr	Table 10C — after 30-day SOP gate
Gujarat — total to implement	₹38.4 Cr	₹32 Cr build + ₹6.4 Cr Year-1 hold · Table 10H
Gujarat — time to implement	36 months	Go-live; first city at month 6 · Table 10J

Table 12. Two wallets: students spend nothing; the scale plan is a government product budget.

11. Team, time, and work packages

Event window: challenge expected to start September 2026; Stage 1 open innovation (students & MSME vs large tech); top six to live finale. We work as student category.

Role	Work package
Lead / architect	HLD, model choice Hybrid, mentor comms, submission pack
Backend	Registry, adapters, hash chain, watchlist API, CSV onboard
AI	YOLOv8+track, ANPR, match score, clip crop
Frontend + GIS	React mosaic, Leaflet map, alert inbox, route polyline
Cyber + report	Tamper demos, AES pack, DPDP/ethics, output report, video

Table 13. Five hats; 4 students can double-up (AI+backend, lead+cyber).

Calendar if we start immediately (mid-August 2026)

Week	Output
W1	This mentor sign-off; repo; schema; dummy GIS of 50 Gujarat points
W2	Ingest 4 sources + hash chain + tamper button
W3	ANPR on own clips + watchlist match + alerts
W4	GIS route + searchable events + CSV 50-cam onboard
W5	HLD polish, PPT, 6-min video, ethics page; mock “find this plate” drill
Sept+	Portal submit; then harden for government-feed / finale

Table 14. Five focused weeks to a complete Step-5 pack (evenings + weekends).

12. Ethics, law, and what we will not claim

- DPDP Act, 2023: purpose = lawful policing assistance; retention limits; no publishing government faces/plates.
- Face watchlist only if legal cell / mentor allows; otherwise vehicles-only for Stage 1 (still satisfies “designated vehicle”).
- No claim of 99% ANPR or court-certified evidence; hash chain is a technical aid for NFSU-style custody.
- YOLOv8 AGPL: OK for challenge; commercial police production may need licence or model swap — stated in HLD.
- Human confirmation before any public-facing alert.

- Bias: document OCR weakness on dirty / non-Latin / fancy fonts.

13. Gap analysis vs the portal pictures (nothing left implicit)

Portal item	In first college idea?	In this plan?
Heterogeneous / multi-VMS / 80k	No (4 home cams)	Yes — adapters + scale chapter
Model 1 Registry & GIS mandatory	No	Yes — first-class module
Hybrid vs Model 4	Implied central app	Yes — federation chosen
Watchlists (stolen, wanted, missing...)	Partial (face/ANPR toys)	Yes — schema + matcher
All 10 Step-3 tiles	Only AI + some cyber	Yes — Sections 4.1–4.10
50-cam live + GIS route	No	Yes — Table 8
Own-feed + gov-feed + HLD + video	No	Yes — Table 9
DR, bandwidth, retention, rollout	No	Yes — Section 8
Cyber tamper / hash	Yes (CyberShield)	Yes — bonus differentiator
Weapon / deepfake as core	Yes (over-scoped)	Parked; not the official test case

Table 15. We reshaped the project to the official problem, not the other way around.

14. One-paragraph synopsis (for college file / portal abstract)

SentinelShield is a hybrid statewide CCTV integration platform designed for the Gujarat Police Innovation Challenge 2026. It implements the mandatory camera Registry and GIS layer, federates heterogeneous VMS/vendor streams without replacing existing NVRs, and runs AI analytics (vehicle tracking, Indian ANPR, watchlist matching) only on authenticated video. A cybersecurity engine maintains SHA-256 hash chains and detects freeze, blackout and replay so that alerts on tampered feeds are marked untrusted. The prototype is built at zero cash cost on open-source software and is structured to onboard ~50 mixed cameras, find a designated vehicle, show GIS movement history, and scale in design to approximately 80,000 cameras by pushing inference to the edge and keeping raw video with owning departments.

15. Risks if we ignore the official steps

- A beautiful YOLO demo with no GIS/registry will fail Step 2 (Model 1 mandatory) and Step 4.
- Promising a new central VMS (Model 4) will be marked naïve by police IT.
- No watchlist schema ⇒ cannot “identify specified vehicle” under evaluation.
- No own-feed backup ⇒ government VPN fails and we have nothing to show.
- No scale chapter ⇒ weak on Step 6 / “PoC readiness”.

16. Decision requested from the mentor

- Approve project title **SentinelShield** as the college entry for GPIC 2026 / related internal review.
- Approve **Hybrid model (1 + 3 + 2)** with Model 1 Registry & GIS as week-1 work.
- Approve **vehicle watchlist + ANPR + GIS route** as the primary AI (faces only with written ethics OK).
- Approve parking of weapon/deepfake as non-core so we match the live test case.
- Approve **■0** prototype (no purchase).
- Optional: lab GPU slot and a 30-minute mid-review after W3 (first plate match on map).
- Nominate / confirm team roles and whether we register as student category on sentinel.gujarat.gov.in.

Document	SentinelShield Mentor Briefing — aligned to official Steps 1–7
Event	Gujarat Police Innovation Challenge 2026 · sentinel.gujarat.gov.in
Date	18 August 2026
Mentor decision	Approved / Approved with remarks / Revise (circle)

Remarks	
Mentor name, dept., signature, date	
Team (names / enrolment)	

Sources for event facts: official portal step structure (team screenshots, 18 Aug 2026); public reporting on GPIC 2026 (≈80,000 cameras, live feeds, █37 lakh prizes, i-Hub / DA-IICT / NFSU, September start). Infrastructure rupee/GPU figures are indicative design, not vendor quotes. This is an academic briefing, not a certified police product.